FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014: 2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Main), 2015

ANSWERS, HINTS & SOLUTIONS

HALF COURSE TEST – VII
(Main)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A	C	B
2.	C	B	D
3.	B	C	D
4.	B	B	B
5.	D	C	A
6.	C	B	D
7.	D	C	B
8.	B	A	C
9.	A	B	A
10.	D	B	A
11.	C	A	A
12.	C	C	A
13.	B	D	B
14.	B	A	C
15.	A	D	A
16.	A	A	C
17.	B	D	C
18.	B	A	D
19.	C	C	C
20.	A	C	B
21.	C	D	A
22.	C	A	A
23.	C	B	B
24.	D	B	A
25.	A	C	D
26.	A	C	A
27.	C	D	A
28.	B	B	B
29.	D	D	D
30.	B	A	C

Physics

PART – I

1. Initial acceleration $a = \frac{kt}{M+m}$

After some instant upper block slide over the plank and hence acceleration of the plank remains constant.

3. $mgh = \frac{1}{2}mv^2$ . . . (1)

$\frac{1}{2}\lambda mR^2 \times \omega^2 + \frac{1}{2}mv^2 = mgh$

$\Rightarrow v' = \frac{v}{\sqrt{2}}$

4. Change in kinetic energy $\Delta k = \frac{1}{2} \times 8 \times 5 = 20J$

5. Kinetic energy, $k = \frac{1}{2}mv^2$

$v_1 = 2m/s$ and $v_2 = 4m/s$

Now $4 \times 2 - 1 \times 4 = (4+1)v$

$\Rightarrow v = \frac{4}{5}$

$\therefore \quad \frac{\Delta k}{k} \times 100 = \left(\frac{8 \times 9}{16 \times 5}\right) \times 100 = 90\%$

6. $F = -\frac{k}{r^2} = \frac{mv^2}{r}$

$\Rightarrow KE = \frac{1}{2}mv^2 = \frac{-k}{2r}$

and potential energy = $-\int F dr = \int \frac{k}{r^2} dr = \left[-\frac{k}{r}\right]$

7. $\vec{P} = 2\cos kt\left(\hat{i} + \hat{j}\right)$

$\vec{F} = -2k\sin kt\left(\hat{i} + \hat{j}\right)$

Angle between $\vec{F}$ and $\vec{P}$ is 180°

8. Potential energy = $\frac{1}{2}kx^2 = Mgx$

$\Rightarrow x = \frac{2Mg}{k}$

9. $y = mx + c$ and $y = x+k$
$\Rightarrow \theta = 45°$ and $c = k$
$\therefore$ Angular momentum, $L = mv$ $mv \times k \times \frac{1}{\sqrt{2}} = \frac{mvk}{\sqrt{2}}$

10. Angular momentum, $L = I\omega$
$\frac{MR^2}{2}\omega_1 = mR^2\omega_2 \Rightarrow \omega_1 = 2\omega_2$
$\frac{a_1}{a_2} = \frac{\omega_1^2 R}{\omega_2^2 R} = 4:1$

11. $F = kx = ma$
$\Rightarrow a \propto -x$

14. $y = \frac{x^2}{2} = \frac{t^4}{8}$
$v_y = 4$ (At t = 2 sec)
and $x = \frac{t^2}{2}$
$v_x = 2$ (At t = 2 sec)
$\left(2\hat{i} + 4\hat{j}\right)$ m/s

16. From the principle of flotation
$g\left(m_B + m_C\right) = \left(\frac{m_B}{\rho_B} + \frac{m_C}{\rho_C}\right) \times 1 \times g$
$\left(m_B + m_C\right) = \frac{m_B}{2} + \frac{m_C}{0.5}$
$\frac{m_B}{m_C} = \frac{2}{1}$

17. $v_{max} = A\omega = 2\pi fA$
Energy density $E = \frac{1}{2}\rho v_{max}^2$
So the curve between E and v_{max} is a parabola of increasing slope and symmetric about y-axis.

18. $\frac{1}{2}kA^2 = 9.5 = 4J$
$k = \frac{8}{A^2} = \frac{8}{\left(10^{-2}\right)^2} = 8 \times 10^4$ N/m
$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{2.0}{8 \times 10^4}} = 3.14 \times 10^{-2}$ sec

19. From the figure
T = 4 sec

$$\omega = \frac{2\pi}{T} = \frac{\pi}{2} = 1.57\,\text{rad/sec}$$

$$\omega = \sqrt{\frac{k\omega}{m}} \Rightarrow k = (0.8)\left(\frac{\pi}{2}\right)^2 = 1.97\,\text{N/m}$$

$$|a| = \omega^2 x = 0.197\,\text{m/s}^2$$

20. $$6\pi\eta r v_T + \frac{4}{3}\pi r^3 \rho_{air} g = \frac{4}{3}\pi r^3 \rho_\ell g$$

$$v_T = \frac{2r^2(\rho_\ell - \rho_{air})g}{9\eta}$$

Substituting values we get,

$$v_T = 0.21 \text{ cm/s}$$

21. $$T_1 (V_a)^{\gamma-1} = T_2 (V_d)^{\gamma-1}$$

$$\frac{T_1}{T_2} = \left(\frac{V_d}{V_a}\right)^{\gamma-1}$$

Also

$$T_1 (V_b)^{\gamma-1} = T_2 (V_c)^{\gamma-1}$$

$$\frac{T_1}{T_2} = \left(\frac{V_c}{V_b}\right)^{\gamma-1}$$

$$\frac{V_a}{V_d} = \frac{V_b}{V_c}$$

22. $$k_{max} = E - U_{min}$$

=36 – 20 = 16 J

23. $$m_1 = m_2 = \text{volume} \times \text{density}$$

$$F = \frac{Gm_1m_2}{r^2} = \frac{G\left(\frac{4}{3}\pi r^3\right)\left(\frac{4}{3}\pi r^3\right)\rho^2}{r^2}$$

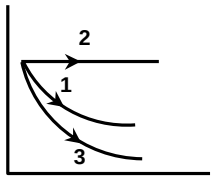
$$F \propto r^4$$

24. $$F = -\int_0^\pi \frac{Gm\left(\frac{M}{L}\right)r d\theta . \sin\theta}{r^2} = \frac{GMm}{Lr}\cos\theta\Big|_0^\pi$$

$$= \frac{GMm}{Lr}(-1) - \frac{GMm}{Lr} = \frac{2GMm}{Lr} = \frac{2GMm\pi}{L^2}$$

25. Area under PV graph = work done

Here area A_2 > Area A_1 > Area A_3



27. Let the thickness of the wall is t

Weight of the sphere = $\frac{4}{3}\pi\left[R^3-(R-t)^3\right]$

Weight of the sphere = $\frac{4}{3}\pi R^3$

For flotation $\frac{4}{3}\pi R^3 > \frac{4\pi\rho}{3}\left[R^3-(R-t)^3\right]$

$t \le \frac{R}{3\rho}$

28. Velocity of liquid from bottom hole $v_1=\sqrt{2g(h+y)}$

Velocity of liquid from top hole $v_2=\sqrt{2gh}$

Force on upper hole = $a\rho v_2^2$

Force on lower hole = $a\rho v_1^2$

Net force on the tank = $a\rho\left(v_1^2-v_2^2\right)=2a\rho gy$

Chemistry

PART – II

4. $V_1 = 5$ litre

$P_{ext} = 1$ atm, $T = 40° + 273 = 313$ K

$$V_2 = \frac{nRT}{P} = \frac{10 \times 0.0821 \times 313}{1} = 256.97 \text{ litre}$$

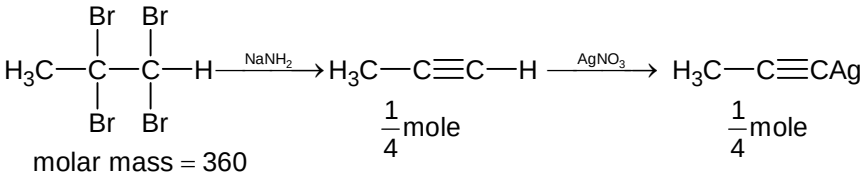
$W = -P_{ext}(V_2 - V_1)$

$= -1(256.97 - 5) = 251.97$ L atm

$= -251.97 \times 101.3$ J

$= -25524.8$ J $= -2.55$ kJ

6.



molar mass $= 360$

$$\text{mole} = \frac{90}{360} = \frac{1}{4}$$

Mole of ppt $= \frac{1}{4}$ mole

Weight $= \frac{1}{4} \times 147 = 36.75$

7. $\underset{n=2}{Na_2CO_3} + 2HCl \longrightarrow 2NaCl + Na_2CO_3$

$\underset{n=1}{NaHCO_3} + HCl \longrightarrow NaCl + CO_2 + H_2O$

$Na_2CO_3 + NaHCO_3$

Wt. x gm $1 - x$

Mole $= \frac{x}{106} = \frac{1-x}{86}$ ...(1) equimolar amount

Meq of Na_2CO_3 + Meq of $NaHCO_3$ = Meq of HCl

$$\frac{x}{106} \times 2 + \frac{1-x}{86} = 0.1 \times V \quad ...(2)$$

From (1)

$$\frac{x}{106} = \frac{1-x}{86}$$

$86x = 106 - 106x$

$192x = 106 \Rightarrow x = 0.552$

$1 - x = 0.448$

From (2)

$$\frac{0.552}{106} \times 2 + \frac{0.448}{86} = 0.1 \times V$$

$0.0104 + 0.00520 = 0.1 \times V$

$0.01562 = 0.1$ V

V = 0.1562 litre

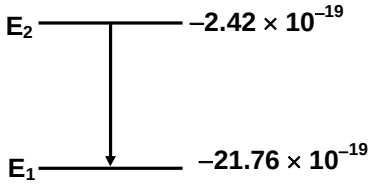
V = 156 ml

8. $3Cl_2 + 2Fe \longrightarrow 2FeCl_3(s)$

moles of $Fe = \frac{1}{56} = 0.01786$

mole of $Cl_2 = \frac{3}{2} \times 0.01786 = 0.02678$

Requirement volume of Cl_2 at STP will = 0.02678×22.4
= 0.6 litre

9. 

$\Delta E = E_2 - E_1$

$= -2.42 \times 10^{-19} + 21.76 \times 10^{-19} = \frac{hc}{\lambda}$

or $19.34 \times 10^{-19} = \frac{6.625 \times 10^{-34}}{\lambda} \times 3 \times 10^8$

$\lambda = \frac{19.875 \times 10^{-26}}{19.34 \times 10^{-19}} = 1.0276 \times 10^{-7}\,m$

= 1027 $\mathring{A}$

10. 2.5 litre O_2 at STP

Moles of $O_2 = \frac{2.5}{22.4} = 0.1116$

For N_2:

PV = nRT

$2.5 \times V = 0.1116 \times 0.0821 \times 573$

V = 2.1 litre

12. $\mu_{exp} = e \times d = \left(4.8 \times 10^{-10}\right) \times \left(2.36 \times 10^{-8}\right)$ e.s.u cm

$= 11.328 \times 10^{-18}$ e.s.u × cm

= 11.328 D

% ionic character = $\frac{\mu_{obs}}{\mu_{exp}} \times 100 = \frac{8.5}{11.328} \times 100$

= 75.03%

13. $Cd(CN)_2 \rightleftharpoons Cd^{2+} + 2CN^-$

S $(2S + 0.50)$

$NaCN \longrightarrow Na^+ + CN^-$

0.50 M

$K_{sp} = S \times (0.05)^2$

$S = \frac{1 \times 10^{-8} \times 100 \times 100}{0.05 \times 0.05}$

$= 4 \times 10^{-6}$

14. $\Delta G^0 = 2\left[\Delta G_f^0 NOBr\right] - 2\left[\Delta G_f^0 NO\right] + \Delta G_f^0\left[Br_2\right]$

$= 2 \times 82.4 - 2 \times 86.55 + 1 \times 0$

$= -8.3$ kJ $= -8.3 \times 10^3$ J

Now, $2NO(g) + Br_2(\ell) \longrightarrow 2NOBr$

$$Q_p = \frac{\left(pNOBr(g)\right)^2}{\left(pNO(g)\right)^2} = \frac{2^2}{(0.1)^2} = 4 \times 10^2$$

$\Delta G = \Delta G^0 + RT \ln Q_p$

$= -8.3 \times 10^3 + (8.314 \times 298) \ln 4 \times 10^2$

$= -8.3 \times 10^3 + 1.5 \times 10^4$ J

$= +6.7 \times 10^3 = 6.7$ kJ

15. $$\ln\frac{K_1}{K_2} = -\frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

$$\ln\left(\frac{1}{3}\right) = -\frac{E_a}{8.314}\left[\frac{1}{298} - \frac{1}{313}\right]$$

$\Rightarrow E_a = 5.7 \times 10^4$ Jmol^{-1}

$= 57$ kJ/mole

16. Since specific heat of solution is not given assuming dilute have the same specific heat capacity as does water.

$q = mc_s\Delta T$

$= 300 \times 4.18 \times (1.96)$

$= 2.46$ kJ

Volume of solution = 100 + 200 ml

= 300

Weight of solution = 300 g

This heat flow is used to calculate ΔH for 1 mole of the reaction.

$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(\ell)$

Moles of HCl reacted $= 0.1 \times 0.44$

$= 0.044$ mole

1 mole HCl yield 1 mole of H^+

0.044 mole HCl yield 0.044 mole of H^+

$H^+ + OH^- \longrightarrow H_2O(\ell)$

Neutralization of 0.044 mole of H^+ generates 2.46 kJ of heat.

The reaction is exothermic:

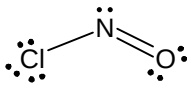
ΔH = –ve

4.4×10^{-2} mole H^+ (aq) = –2.46 kJ

1 mole H^+ (aq) $= -\frac{2.46}{4.4 \times 10^{-2}}$

$= -56$ kJ

$\Delta H_{reaction} = -56$ kJ

18. 

19.

	σ bond pair	Lone pair	Hybridization	Shape
$HgCl_2$	2	0	sp	Linear
$SnCl_2$	2	1	sp^2	V – shape
F_2O	2	2	sp^3	V – shape

20. $\frac{t_x}{t_{O_2}} = \sqrt{\frac{M_x}{M_{O_2}}} \Rightarrow \frac{8.41}{3.25} = \sqrt{\frac{M_x}{M_{O_2}}}$

$\Rightarrow M_x = 214.27$

21. Initially moles of gas at 4.33 atm

$n_1 = \frac{PV}{RT} = \frac{4.33 \times 3.11}{RT}$ ...(1)

Moles of gas in 2.11 L container at 2.55 atm

$n_2 = \frac{2.55 \times 2.11}{RT}$ ...(2)

On adding total moles = $n_1 + n_2$

$\frac{4.33 \times 3.11}{RT} + \frac{2.55 \times 2.11}{RT}$

New pressure in the container of volume 3.11 L

$P = \frac{nRT}{V} = \left[\frac{4.33 \times 3.11 + 2.55 \times 2.11}{RT \times 3.11}\right] RT = 6.06$ atm

22. $\frac{1}{\lambda} = Z^2 R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

$R_H = 1.097 \times 10^7\ m^{-1}$

$Z = 6$ for carbon

$n_1 = 1, n_2 = 3$

$\frac{1}{\lambda} = 1.097 \times 10^7 \times (6)^2 \left[\frac{1}{1^2} - \frac{1}{3^2}\right]$

$= 1.097 \times 10^7 \times 36 \left[\frac{9-1}{9}\right]$

$= 1.097 \times 10^7 \times 36 \times \frac{8}{9} = 35.1 \times 10^7 = 2.85 \times 10^{-9} = 2.85$ nm

24. $CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(\ell)$

$\Delta H = -890.3$ kJ

1 mole CH_4 produce = -890.3 kJ

$\Rightarrow -890.3$ kJ heat is produced by 1 mole CH_4

1 kJ heat is produced by $\frac{1}{890.3}$ mole CH_4

356.4 kJ heat is produced by $\frac{356.4}{890.3} = 0.4$ mole

Weight of CH_4 required = $0.4 \times 16 = 6.4$ gm

25.

	N_2	+	$3H_2$	$\longrightarrow$	$2NH_3$
mole	$\frac{100}{28}$				$\frac{100}{28} \times 2$

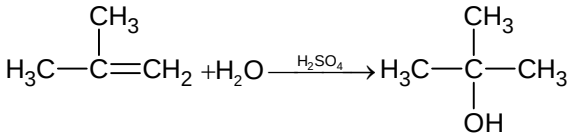
$4NH_3 + 5O_2 \longrightarrow 4NO + 6H_2O$

4 mole NH_3 gives 4 mole of NO

$\frac{200}{28}$ mole NH_3 gives 4 moles of $\frac{200}{28}$ mole of NO

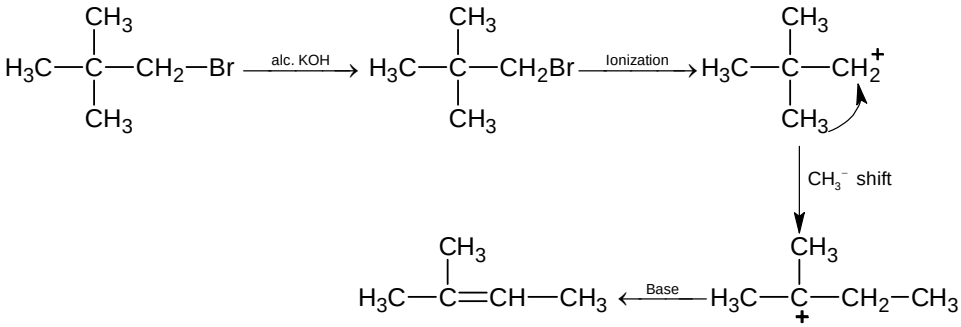
Moles of NO formed from 100 g N_2 = $\frac{200}{28}$

Weight of NO formed = $\frac{200}{28} \times 30\text{gm} = 214.28$ gm

26. 

Optically inactive

27. During electrolysis, two molecule of potassium salt of mono carboxylic acid combine to form alkane along with the lost of two COO^- group. Hence alkane formed will contain = 2n – 2 (no of C–atom).

28. 

29. Adding inert gas at constant volume does not affect any equilibrium whether $\Delta n \geq 0$ or $\Delta n \leq 0$.

30. BF_3 is least acidic among boron halide.

Mathematics PART – III

1. $(a, b) \in R$ if a and $b \in A_j$
$\Rightarrow (b, a) \in R$ which gives symmetry
$(a, a) \in R$ gives reflexivity
Similar argument gives transitivity also max value of $n(R) = 15^2$

2. The given condition suggest that a lies between the roots. Let $f(x) = 2x^2 - 2(2a + 1)x + a(a + 1)$ for a to lie between the roots, $f(a) < 0$
$2a^2 - 2a(2a + 1) + a(a + 1) < 0$
$-a^2 - a < 0$
$a^2 + a > 0$
$a > 0$ or $a < -1$

3. $\sin^6\theta + \cos^6\theta + k\cos^2 2\theta = 1$
$(\sin^2\theta + \cos^2\theta)(\sin^4\theta - \sin^2\theta\cos^2\theta + \cos^4\theta) + k\cos^2 2\theta = 1$
$\sin^4\theta + \cos^4\theta - \sin^2\theta\cos^2\theta + k\cos^2 2\theta = 1$
$(\sin^2\theta + \cos^2\theta)^2 - 3\sin^2\theta\cos^2\theta + k\cos^2 2\theta = 1$
$3\sin^2\theta\cos^2\theta = k\cos^2 2\theta$
$\frac{3}{4}\sin^2 2\theta = k\cos^2 2\theta$
$k = \frac{3}{4}\tan^2 2\theta$

4. $\tan x = \lambda$, $\tan y = 2\lambda$, $\tan z = 3\lambda$
$x + y + z = \pi$
$\tan x + \tan y + \tan z = \tan x \tan y \tan z$
$6\lambda = 6\lambda^2 \Rightarrow 6\lambda(\lambda^2 - 1) = 0$
$\lambda = \pm 1$, $(\lambda \neq 0)$
$\tan x = \pm 1$, $\tan y = \pm 2$, $\tan z = \pm 3$
$\tan x + \tan y + \tan z = \pm 6$
Maximum value 6 minimum values is –6

5. Let $f(x) = ax^2 + 25x - 3c$
Since $f(x) = 0$ has non real roots, $f(x)$ will have the same sign for all values of x
Given $\frac{3c}{4} < (a + b)$
$\Rightarrow 4a + 4b - 3c > 0$
$f(2) > 0$
$f(x) > 0 \ \forall\ x \in R$
$f(0) > 0$
$\Rightarrow c < 0$

6. (A) $f(x) = x^4 + 2x^3 - x^2 + 1 \rightarrow$ A polynomial of degree even will always be into say
$f(x) = a_0x^{2n} + a_1x^{2n-1} + a_2x^{2n-2} + ... + a_{2n}$
$\underset{x\to\pm\infty}{\text{Limit}}\ f(x) = \underset{x\to\pm\infty}{\text{Limit}}\ [x^{2n}\left(a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + ... + \frac{a_{2n}}{x^{2n}}\right) = \begin{cases} \infty & \text{if } a_0 > 0 \\ -\infty & \text{if } a_0 < 0 \end{cases}$
Hence it will never approach $\infty / -\infty$
(B) $f(x) = x^3 + x + 1 \Rightarrow f'(x) = 3x^2 + 1$ - injective as well as surjective

(C) $f(x) = \sqrt{1+x^2} -$ neither injective nor surjective (minimum value = 1)

$f(x) = x^3 + 2x^2 - x + 1 \Rightarrow f'(x) = 3x^2 + 4x - 1 \Rightarrow D > 0$

Hence f(x) is surjective but not injective

7. Given equation of line is $\frac{ax}{c-1} + \frac{by}{c-1} + 1 = 0$

Line contains two independent parameters $\frac{a}{c-1}$ and $\frac{b}{c-1}$. A line will pass through a fixed point if it contains only one independent parameter

$\Rightarrow 5a + 4b + 20c = t$

$\frac{5a}{c-1} + \frac{4b}{c-1} = \frac{t-20c}{c-1}$

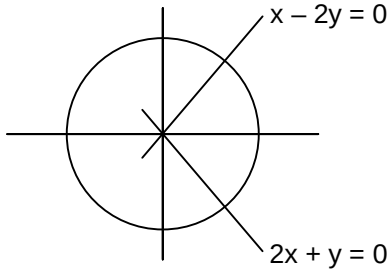
In this equation R.H.S will the independent of c if t = 20. Then we can express $\frac{b}{c-1}$ in terms of $\frac{a}{c-1}$ and vice-versa. Hence, if t = 20 then line will pass through a fixed point

8. $2x^2 - 3xy - 2y^2 = 0$

$(x - 2y)(2x + y) = 0$

Which represents a pair of perpendicular lines passing through the origin

Required area = $\frac{\pi(2)^2}{4} - \frac{\pi(1)^2}{4} = \frac{3\pi}{4}$ sq. units



9. As $x = |y| > 0$

Circle lies in I and IV quadrant

Centre of circle lies on x-axis

$OC = \sqrt{(OA)^2 + (AC)^2} = \sqrt{2+2} = 2$

Equation of circle is $(x - 2)^2 + (y - 0)^2 = 2$

$x^2 + y^2 - 4x + 2 = 0$

10. The point A is $(a\cos\alpha, 2 + a\sin\alpha)$

Since the triangle ABC is equilateral its circum centre (0, 2) coincides with its centroid

$\left(\frac{1}{3}\sum\cos\alpha,\ \frac{1}{3}\sum\sin\alpha\right) = (0,\ 2)$

$\sum\cos\alpha = 0$ and $\sum\sin\alpha = 6$

11. $2x^2 - xy - y^2 + x + 2y - 1 = 0$

$(2x + y - 1)(x - y + 1) = 0$

So the sides $x - y + 1 = 0$ and $x + y + 1 = 0$ are $\perp$ to each other the orthocentre is intersection of $\perp$ sides

12. $\sqrt{(x-2)^2 + y^2} = 4 - \sqrt{(x+2)^2 + y^2}$ ..... (1)

But $(x + 2)^2 + y^2 - [(x - 2)^2 + y^2] = 8$ ..... (2)

Dividing (2) by (1) we get

$\sqrt{(x+2)^2 + y^2} - \sqrt{(x-2)^2 + y^2} = 2x$ ..... (3)

Adding (1) and (2) we get

$2\sqrt{(x+2)^2+y^2} = 4+2x$

$(x+2)^2+y^2 = (x+2)^2$

$y^2 > 0$

Pair of coincident lines

13. Polar of (x_1, y_1) w.r.t. parabola $y^2 = 4a$ is $yy_1 = 2a(x + x_1)$

$2ax - yy_1 + 2ax_1 = 0$

Compare this with $\ell x + my + n = 0$

$\frac{2a}{\ell} = -\frac{y_1}{m} = \frac{2ax_1}{n} \Rightarrow x_1 = \frac{n}{\ell}$

$y_1 = -\frac{2am}{\ell}$ pole $\left(\frac{h}{\ell}, -\frac{2am}{\ell}\right)$

14. $\frac{x}{a}+\frac{y}{b} = \sqrt{2}$ is tangent to $\frac{x^2}{a^2}+\frac{y^2}{b^2} = 1$ ..... (1)

at P(a cos θ, b sin θ)

$\frac{xx_1}{a^2}+\frac{yy_1}{b^2} = 1$

$\frac{xa\cos\theta}{a^2}+\frac{by\sin\theta}{b^2} = 1$

$\frac{x\cos\theta}{a}+\frac{y\sin\theta}{b} = 1$ ..... (2)

$\frac{x}{a\sqrt{2}}+\frac{y}{b\sqrt{2}} = 1$

$\cos\theta = \sin\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{\pi}{4}$

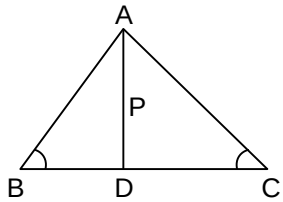
15. When both are ellipse, then e, e′ < 1 and so $e^2 + e'^2 = 3$ is not true and so it false

When both are parabolas $e = 1 = e' \Rightarrow e^2 + e'^2 = 3$ not true

When both are hyperbolas $e^2 + e'^2 = 3$ possible e, e′ > 1

16. $BD = P\cot 22\frac{1}{2}^o$

$DC = P\cot 112\frac{1}{2}^o$



$\text{Base} = P\left[\frac{\cos 22\frac{1}{2}^o}{\sin 22\frac{1}{2}^o}+\frac{\cos 112\frac{1}{2}^o}{\sin 112\frac{1}{2}^o}\right]$

$= P\frac{2\sin 135^o}{2\sin 22\frac{1}{2}^o \sin 112\frac{1}{2}^o} = \frac{2P\frac{1}{\sqrt{2}}}{\cos 90^o - \cos 135^o} = 2P$

$P = \frac{\text{Base}}{2}$

17. $\sum \frac{\cot A}{a} = \sum \frac{b^2+c^2-a^2}{2abc} = \frac{a^2+b^2+c^2}{2abc} = \frac{(a+b+c)^2-2\Sigma ab}{2abc} = \frac{121-76}{80} = \frac{9}{16}$

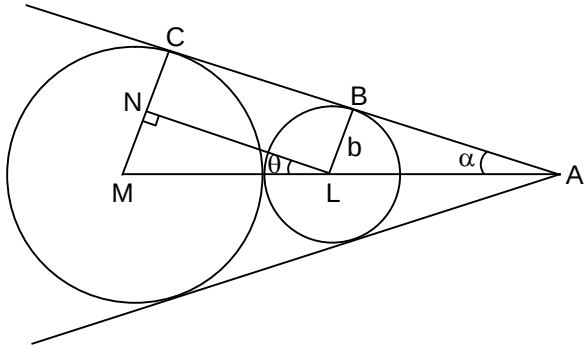
18. From ΔMLN

$\sin\alpha = \frac{a-b}{a+b}$

$\alpha = \sin^{-1}\frac{a-b}{a+b}$

Angle between AB and AD

$2\alpha = \sin^{-1}\left(\frac{a-b}{a+b}\right)$



19. Tangents to parabola $y^2 = 4x$ having slope m is $y = mx + \frac{1}{m}$ tangents to circle having slope m is

$y + 2 = m(x - 1) + 4\sqrt{1+m^2}$

Distance between tangents = $\left|\frac{4\sqrt{1+m^2} - m - 2 - \frac{1}{m}}{\sqrt{1+m^2}}\right|$

$= \left|4 - \frac{2}{\sqrt{1+m^2}} - \frac{\sqrt{m^2+1}}{m}\right| \Rightarrow m > 0$

Hence $d < 4$

20. Let the ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

A = (a cos θ, b sin θ)

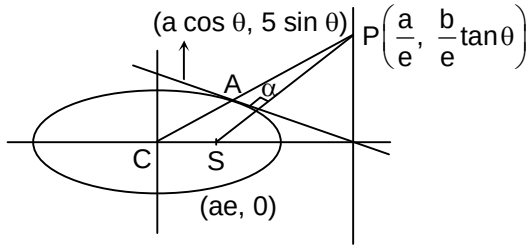
Equation of AC will be $y = \frac{b}{a}\tan\theta\, x$

Solving with $x = \frac{a}{e}$, we get $P\left(\frac{a}{e}, \frac{b}{e}\tan\theta\right)$

Slope of tangents at A is $-\frac{b}{a\tan\theta}$

Slope of $PS = \frac{\frac{b}{e}\tan\theta}{\frac{a}{e} - ae} = \frac{b\tan\theta}{a(1-e^2)} = \frac{a}{b}\tan\theta$

$\Rightarrow \alpha = \frac{\pi}{2}$



21. D should be $b^2 - 4ac > 0$

22. Using A.M. ≥ G.M.

$\frac{\sin A + \frac{1}{\sin A} + 1}{3} \geq 1$

$\sin A + \frac{1}{\sin A} + 1 \geq 3$

Similarly $\sin B + \frac{1}{\sin B} + 1 \geq 3$

$\sin C + \frac{1}{\sin C} + 1 \geq 3$

$\sum\left(\sin A + \frac{1}{\sin A} + 1\right) \geq 9$

23. $AL_1^2 + L_1M_1^2 = (a^2 + 1^2) + \left[(a-1)^2 + 1^2\right]$

$AL_2^2 + L_2M_2^2 = (a^2 + 2^2) + \left[(a-2)^2 + 2^2\right]$

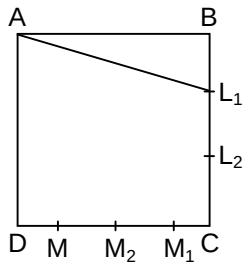
$AL_{a-1}^2 + L_{a-1}M_{a-1}^2 = a^2 + (a-1)^2 + \left[1^2 (a-1)^2\right]$

The required sum

$= (a-1)a^2 + \left[1^2 + 2^2 + ...(a-1)^2\right] + 2\left[1^2 + 2^2 + ...(a-1)^2\right]$

$(a-1)a^2 + \frac{3(a-1)a(2a-1)}{6}$

$\Rightarrow a(a-1)\left[a + \frac{2a-1}{2}\right]$



24. $\frac{1}{\sqrt{a+rx} + \sqrt{a+(r-1)x}} = \frac{1}{x}\left[\sqrt{a+rx} - \sqrt{a+(r-1)x}\right]$

$t_1 + t_2 + ... + t_n = \frac{1}{x}\left[\sqrt{a+nx} - \sqrt{a}\right] = \frac{n}{\sqrt{a} + \sqrt{a+nx}}$

25. $\alpha + \beta = -a$

$\alpha\beta = b$ ..... (1)

$\alpha^4 + \beta^4 = p$

$\alpha^4\beta^4 = q$ ..... (2)

Using (1) and (2), we get $(a^2 - 2b)^2 = p + 2b^2$

$a^2 - 4ba^2 + 2b^2 - p = 0$

a^2 is one of roots of $x^2 - 4bx + 2b^2 - p = 0$

Product of roots of equation $= 2b^2 - p$

$2\alpha^2\beta^2 - (\alpha^4 + \beta^4) = -(\alpha^2 - \beta^2) < 0$

Hence one root is positive and other is negative

26. $\log_x 2 \cdot \log_{2x} 2 = \log_{4x} 2$

Here $x > 0$ and $x \neq 1, \frac{1}{2}, \frac{1}{4}$

$\frac{1}{\log_2 x} \cdot \frac{1}{\log_2 2x} = \frac{1}{\log_2 4x}$

$\log_2 x(1 + \log_2 x) = (2 + \log_2 x)$

Put $\log_2 x = t$

$t(1 + t) = 2 + t$

Hence $x = \left\{2^{-\sqrt{2}}, 2^{\sqrt{2}}\right\}$

27. Foci $(\pm 5, 0)$ where $ae = 5$

$$x = \frac{a}{e} \Rightarrow \frac{a}{e} = \frac{36}{5}$$

$$ae \cdot \frac{a}{e} = 5 \cdot \frac{36}{5}$$

$a = 6$

$$e = \frac{5}{6}$$

$b^2 = a^2(1 - e^2) = 11$

28. For $x = 1$

$\sin^7 y = \sec^2 2y + \cos^4 y$

$\sin^7 y \cos^2 2y = 1 + \cos^4 y \cos^2 2y$

RHS ≥ 1 and LHS ≤ 1

LHS $= 1$

$\sin^7 y \cos^2 2y = 1$

$$\sin^7 y = 1 \text{ and } \cos^2 2y = 1 \Rightarrow y = \frac{\pi}{2}$$

General value of y is $2n\pi + \frac{\pi}{2}$

29. $x + 3y - 2 = 0$

Any perpendicular line $3x - y + c = 0$

$y = 3x + c$

Tangents to the hyperbola $\frac{x^2}{1} - \frac{y^2}{3} = 1$

$c^2 = a^2m^2 - b^2 \Rightarrow c = \pm\sqrt{6}$

$y = 3x \pm \sqrt{6}$

30. $f(x)$ is defined when $x^2 - 1 > 0$ and $3 + x > 0$ and $3 + x \neq 1$

$x^2 > 1$ and $x > -3$ and $x \neq -2$

$x < -1$ or $x > 1$ and $x > -3$ and $x \neq -2$

$D_f = (-3, -2) \cup (-2, -1) \cup (1, \infty)$